

# ECON 470 Homework 5

Submission 2, Spring 2026

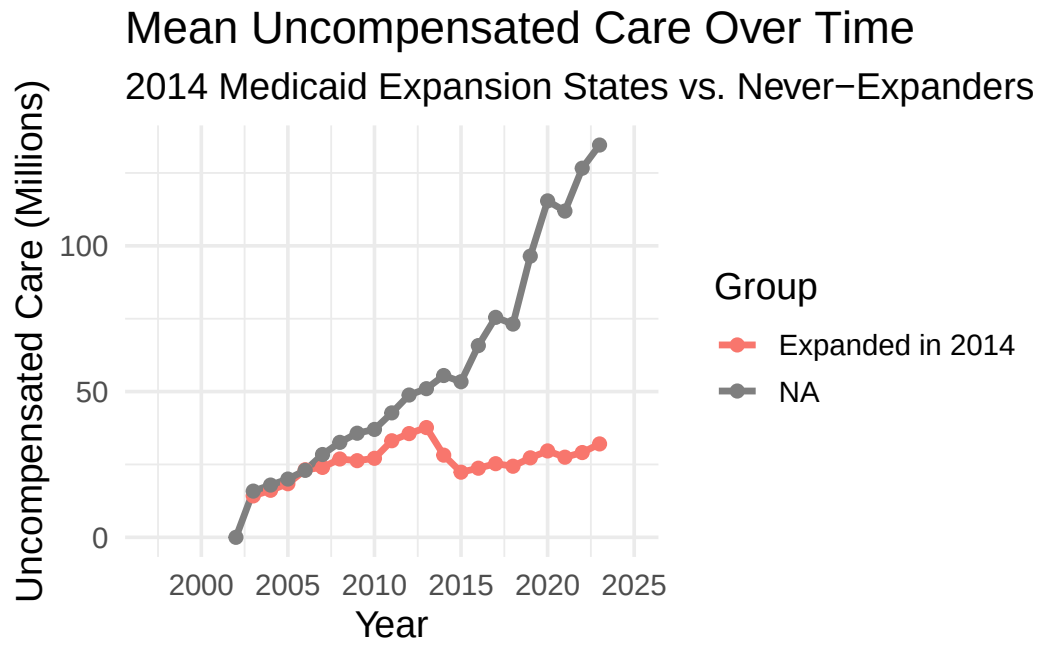
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## Problem 1

Table 1: Uncompensated Care (\$M), 2010–2018

| Year | Mean (\$M) | Std. Dev. (\$M) |
|------|------------|-----------------|
| 2010 | 30.26      | 72.26           |
| 2011 | 34.10      | 74.96           |
| 2012 | 37.54      | 86.63           |
| 2013 | 39.43      | 80.86           |
| 2014 | 36.63      | 88.53           |
| 2015 | 33.37      | 86.90           |
| 2016 | 45.24      | 403.12          |
| 2017 | 41.14      | 102.67          |
| 2018 | 39.10      | 100.40          |

## Problem 2



### Problem 3

Table 2: Difference-in-Differences: Mean Uncompensated Care (\$M)

| Group     | 2012 (\$M) | 2015 (\$M) | Difference (\$M) |
|-----------|------------|------------|------------------|
| Control   | 48.81      | 53.38      | 4.57             |
| Treatment | 35.58      | 22.35      | -13.24           |

#### **Problem 4**

The adoption of Medicaid leads to a decrease in uncompensated care, as observed. Thus, we expect this effect as it encourages hospitals to take Medicare patients and get compensation or partial support from the Medicare program, instead of outsourcing debt to manage or counter their listed charges to a third-party bank.

## Estimate ATEs

### Problem 5

Table 3: DiD Estimates: Effect of Medicaid Expansion on Uncompensated Care (\$M)

| Term                       | Estimate       | Std. Error   | t-Statistic   | p-Value             |
|----------------------------|----------------|--------------|---------------|---------------------|
| Intercept                  | 47.976         | 2.104        | 22.799        | 3.87e-113 ***       |
| Post 2014                  | 15.938         | 2.677        | 5.953         | 2.70e-09 ***        |
| Ever Expanded              | -12.419        | 2.653        | -4.682        | 2.87e-06 ***        |
| Post x Ever Expanded (DiD) | <b>-26.681</b> | <b>3.382</b> | <b>-7.888</b> | <b>3.27e-15 ***</b> |

Observations: 14990

R-squared: 0.024

Signif. codes: \*\*\* p<0.01, \*\* p<0.05, \* p<0.10

## Problem 6

Table 4: TWFE Estimates: Effect of Medicaid Expansion on Uncompensated Care (\$M)

| Term                   | Estimate | Std. Error | t-Statistic | p-Value      |
|------------------------|----------|------------|-------------|--------------|
| <b>Treatment (DiD)</b> | -30.755  | 5.903      | -5.21       | 7.36e-06 *** |

Hospital and year fixed effects included.

Standard errors clustered at the state level.

Observations: 14990

R-squared (within): 0.039

Signif. codes: \*\*\* p<0.01, \*\* p<0.05, \* p<0.10

We notice that both the estimators yield near about same magnitude of estimated effect around 48 to 52 millions of uncompensated care.

## Problem 7

Table 5: OLS DiD: Effect of Medicaid Expansion on Uncompensated Care (\$M), All States

| Term                              | Estimate       | Std. Error   | t-Statistic   | p-Value             |
|-----------------------------------|----------------|--------------|---------------|---------------------|
| <b>Intercept</b>                  | 47.976         | 3.643        | 13.171        | 1.88e-39 ***        |
| <b>Post 2014</b>                  | 15.938         | 4.635        | 3.439         | 5.85e-04 ***        |
| <b>Ever Expanded</b>              | -14.916        | 4.292        | -3.475        | 5.11e-04 ***        |
| <b>Post x Ever Expanded (DiD)</b> | <b>-19.504</b> | <b>5.461</b> | <b>-3.572</b> | <b>3.56e-04 ***</b> |

Observations: 20095

R-squared: 0.006

Signif. codes: \*\*\* p<0.01, \*\* p<0.05, \* p<0.10

Table 6: TWFE: Effect of Medicaid Expansion on Uncompensated Care (\$M), All States

| Term                   | Estimate | Std. Error | t-Statistic | p-Value      |
|------------------------|----------|------------|-------------|--------------|
| <b>Treatment (DiD)</b> | -23.609  | 4.736      | -4.985      | 7.54e-06 *** |

Hospital and year fixed effects included.

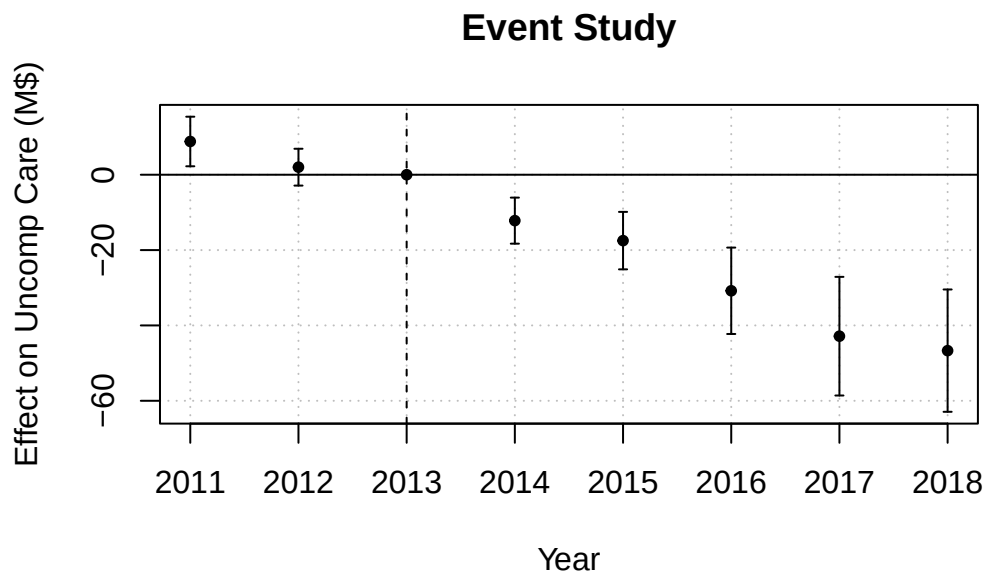
Standard errors clustered at the state level.

Observations: 20095

R-squared (within): 0.002

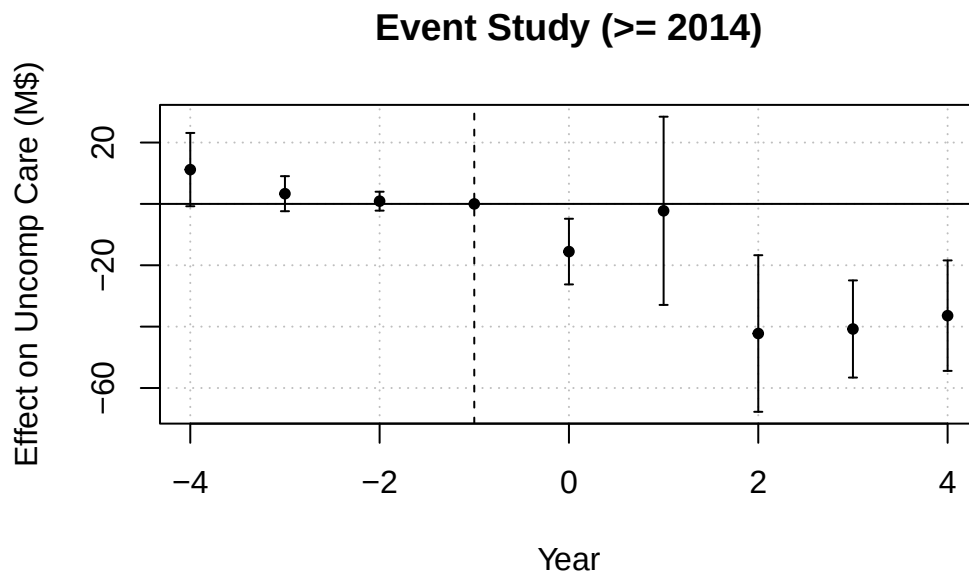
Signif. codes: \*\*\* p<0.01, \*\* p<0.05, \* p<0.10

## Problem 8





Problem 9



## Problem 10

Medicaid expansion leads to a decrease in uncompensated care for hospitals in states that have adopted the change in policy. The event study graph is concerning because the bars for pre-treatment inch away from the zero margin, showing a violation of the parallel trends assumption. One primary limitation of the TWFE estimator is its biased nature in the context of staggered Medicaid expansion, as it doesn't take into account uncha